Murat Ambarkutuk

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Objective

A PhD candidate in Computer Engineering at Virginia Tech, focusing on Machine and Software Intelligence, with a dedication to applying expertise in Computer Vision, Sensor Fusion, and Machine Learning to address complex challenges in robotics, perception, and autonomy. Currently on a J1 Exchange Scholar visa with 36 months of pre-approved work authorization, available to start in mid-September 2024.

Technical & Language Skills

Programming Languages: Python, C++, MATLAB, C, HTML/CSS, JavaScript, LaTeX
Robotics & Machine Learning: ROS, TensorFlow, PyTorch, Keras, scikit-learn, OpenCV, Jupyter, PyMC3
Cloud, DevOps & Infrastructure: Linux, Version Control (Git), Containerization and Orchestration (Docker, Docker Compose, Kubernetes), AWS (EC2, S3, Lambda, SageMaker), CI/CD (Jenkins, GitHub Actions, GitLab CI), Terraform, Ansible
Frameworks & Libraries: numpy, scipy, pandas, Flask, Bootstrap
Languages: Turkish (Native), English (Proficient), German (Novice)

Education

PhD in Computer Engineering — 2018 – September 2024 (Scheduled)

The Bradley Department of Electrical and Computer Engineering at Virginia Tech — *Blacksburg, VA*

- Paul E. Torgersen Research Excellence Award
- The Outstanding Research Award and Fellowship from the Turkish Ministry of National Education

Dissertation: A Sensor Fusion Technique for Gait Analysis from Structural Vibration and Computer Vision

MS in Mechanical Engineering — 2015 – 2018

Mechanical Engineering at Virginia Tech — *Blacksburg, VA*

Thesis: A Grid based Indoor Radiolocation Technique Based on Spatially Coherent Path Loss Model

BS in Mechatronics Engineering — 2008 – 2013

Mechatronics Engineering Department at Kocaeli University — *Kocaeli, Turkey*

Thesis: Box Dimensioning System with Kinect Sensor

Relevant Coursework

- Robust Estimation & Filtering
- Computer Vision
- Advanced Computer Vision
- Advanced Machine Learning
- Bayesian Robotics
- Experimental Robotics
- Exp. Methods & Signal Processing
- Advanced Robotics & Automation

Relevant Professional Experience

Baykar Defense (TR) - Atlas Imaging (US) — 2018 – 2020

Computer Vision Engineer developing orthorechtified maps with aerial photos taken by on autonomous UAVs. — *Remote*

- Derived and actualize pose information (location+orientation) of an UAV and its components based on the distributed measurements of the Inertial Navigation System. These measurements are then used to in an iterative Computer Vision algorithm to project the corners of the images to the ground while considering the elevation changes in the scene.
- Developed a Sensor Fusion algorithm to correct results of the previous stages of the map-making algorithm with statistical representation.
- Implemented Test-driven Development paradigm in the dev-team. On that front, generated tests to check coverage, style requirements, and performance & accuracy of the developed algorithms.
- Architected the CI/CD pipeline to automate the aforementioned tests and generated automated reports from the results of the tests.

Relevant Projects

Sensor Fusion for Occupant Localization in Smart Buildings

Advisors: P Plassmann, CF Jones | Related publications: [5, 4, 2, 1]

- Developed a multi-sensor localization framework that can determine the location of occupants from the measurements of a vibrating floor in the building.
- Proposed a novel sensor fusion algorithm to discard "byzantine" sensors in the sensor network.
- Proposed a novel vibration-based localization model that is the current the state-of-the art in the field.
- Implemented a probabilistic framework to quantify the uncertainty in the localization results.
- Validated the efficacy of the algorithm with controlled experiments and simulations.

Quantification of Collaboration and Communication among Students in a Smart Classroom

Advisors: T Baird, PA Tarazaga | News Article | Project Website

- Designed a sensor-suite (frontend + backend), a multiple-view camera-based system, for a smart classroom to quantify the collaboration among students. The backend creates many functionalities of the suite, i.e., controlling the sensor-suite, scheduling auto-start/stop times as well as employing automated Computer Vision techniques on the collected data. The frontend, on the other hand, exposes the functionality of the backend to remotely control and monitor the system.

Visual Localization of A Car During a Crash Test

Advisor: T Furukawa | Sponsor: Honda Japan

- Designed and manufactured a sensor-suite that provides inertial measurements and images of a downward facing camera to localize a vehicle under the heavy vibrations during car-crash tests.
- Developed an efficient algorithm to localize the vehicle with a combinatoral method that is based on a featured-based localization, visual odometry technique and inertial measurements.

A Grid-based Indoor Radiolocation Technique based on Spatially Coherent Path Loss Model

Advisor: T Furukawa | Publications: [3]

- Derived a probablistic framework of a passive radio-localization technique under multipath propagation that can determine the location of different entities, namely, occupants, robots, and invenvorty, in indoor enviroments.
- Developed the localization algorithm based on the probablistic framework.
- Validated the efficacy of the algorithm with controlled experiments whose results were presented at IEEE RAS conference.

Community Service & Leadership

Graduate Student Ambassador
The Bradley Department of Electrical and Computer Eng.

IEEE Robotics and Automation Society
Student Member & Reviewer

Society of Experimental Mechanics
Student Member

Turkish Student Assoc. at Virginia Tech
Secretary

Relevant Publications

[1] S. Alajlouni, **Ambarkutuk, Murat**, and P. Tarazaga. Towards the detection and localization of multiple occupant footsteps from vibroacoustic measurements. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 38th IMAC, A Conference and Exposition on Structural Dynamics 2020*, page 193. Springer Nature, 2020.

[2] S. Alajlouni, V. V. S. Malladi, and **Ambarkutuk, Murat**. Investigation of using log-spectrum averaging (cepstral averaging) for blind reconstruction of an unknown impact input force. Springer Nature, 2022.

[3] **Ambarkutuk, Murat** and T. Furukawa. A grid-based indoor radiolocation technique based on spatially coherent path loss model. In *2017 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, pages 220–226. IEEE, 2017.

[4] **Ambarkutuk, Murat** and P. Tarazaga. Uncertainty analysis of an occupant localization technique based on simulated structural vibrations. Springer Nature, 2022.

[5] **Ambarkutuk, Murat**, S. Alajlouni, P. A. Tarazaga, and P. E. Plassmann. A multi-sensor stochastic energy-based vibro-localization technique with byzantine sensor elimination. *Sensors*, 23(23), 2023. ISSN 1424-8220. doi: 10.3390/s23239309. URL https://www.mdpi.com/1424-8220/23/23/9309.